

Homework 5: Space–Time finite element method for solving the wave equation

Due by 5/12/2026

1. Consider the one-dimensional wave equation in space–time form:

$$u_{tt} - cu_{xx} = f, \quad (x, t) \in Q := (0, L) \times (0, T),$$

where $c > 0$ is a constant.

The initial conditions are

$$u(x, 0) = u_0(x), \quad u_t(x, 0) = v_0(x), \quad 0 < x < L,$$

and the boundary conditions are

$$u(0, t) = g_0(t), \quad u(L, t) = g_L(t), \quad 0 < t < T.$$

We approximate the solution using continuous piecewise linear finite elements on a triangular mesh of the space–time domain Q . Let

$$u_h(x, t) = \sum_{j=1}^N U_j \phi_j(x, t),$$

where $\{\phi_j\}_{j=1}^N$ are the nodal basis functions.

The general steps are:

- (1) Derive the weak formulation by multiplying by a test function $w(x, t)$, integrate over Q , and integrate by parts in both t and x .
- (2) impose BCs and ICs in the weak formulation.
- (3) Let K be a triangular element with vertices

$$(x_1, t_1), \quad (x_2, t_2), \quad (x_3, t_3).$$

For linear P_1 basis, we have:

$$\phi_i(x, t) = a_i + b_i x + d_i t, \quad i = 1, 2, 3.$$

Find the coefficients a_i, b_i, d_i .

- (4) To derive the local element matrix, you can follow the steps below:

For the elementary triangular element K , define

$$A_{ij}^{(K)} = - \int_K (\phi_j)_t (\phi_i)_t dx dt + c \int_K (\phi_j)_x (\phi_i)_x dx dt.$$

Since ϕ_i is linear on K , its derivatives are constant. Show that

$$A_{ij}^{(K)} = -\Delta K d_j d_i + c \Delta K b_j b_i,$$

where

$$b_i = (\phi_i)_x, \quad d_i = (\phi_i)_t,$$

and ΔK is the area of the triangle.

- (5) Compute the local load vector.

the local load vector is:

$$F_i^{(K)} = \int_K f \phi_i dx dt.$$

You can evaluate this approximately using a quadrature rule.

- (6) Assemble the local matrices $A^{(K)}$ and local vectors $F^{(K)}$ into the global system

$$AU = F + BCs,$$

where BCs contains the contributions from initial and boundary terms.

Follow the steps I listed above, and solve the hyperbolic problem with manufactured solution.

Let $L = 1$, $T = 1$, $c = 1$, and choose the exact solution to be:

$$u_{\text{exact}}(x, t) = \sin(\pi x) \cos(t).$$

- (1) Compute the source term $f(x, t)$ such that

$$u_{tt} - u_{xx} = f.$$

- (2) Compute the initial displacement

$$u_0(x) = u_{\text{exact}}(x, 0).$$

- (3) Compute the initial velocity

$$v_0(x) = \partial_t u_{\text{exact}}(x, 0).$$

- (4) Compute the boundary data

$$g_0(t) = u_{\text{exact}}(0, t), \quad g_L(t) = u_{\text{exact}}(1, t).$$

- (5) Using the space-time finite element method to solve the PDE numerically. Write the details about how you calculate the elementary matrices, how you generate the connectivity, and assemble the global matrix. You can set the spatial and temporal grid to be $N = 6$ and $M = 4$ respectively.
- (6) Bonus questions: Compare the numerical solution u_h with the exact solution u_{exact} at the mesh nodes. Report the maximum nodal error:

$$\max_j |U_j - u_{\text{exact}}(x_j, t_j)|.$$

- (7) Bonus questions: Repeat the computation on at least two refined meshes $N = 12, M = 8$. Does the error decrease as the mesh is refined? Briefly explain your observation.